

Chapter 1

1 Define cloud Computing. Explain NIST definition and essential characteristics.

→ Cloud Computing is a model for delivering of computing services (servers, storage, databases, networking, software) over the internet.

According to the National Institute of Standards and Technology (NIST) definition, "Cloud Computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction."

NIST Essential Characteristics

1 On-demand self-service

- Users can provision resources automatically
- Example:
 - AWS EC2 → Self-provisioning

2 Broad network access

- Services available over the network via standard mechanisms
- Example:
 - Google Cloud Storage → network-based access

3 Resource pooling

- Provider's resources are shared dynamically among users (multi-tenancy)
- Example:
 - Microsoft Azure → resource pooling

4 Rapid elasticity

- Scale resources up/down automatically with demand
- Example:
 - Netflix scales during peak hours

5 Measured Service

- Resource usage is monitored, controlled, and billed per use
- Example:
 - AWS bills hourly or per second

Q2 Explain the need for cloud computing with limitations of traditional IT systems.

→ Traditional IT systems are based on on-premises infrastructure, where organizations manage their own hardware, software, and networking. As business requirements grow and technology evolves rapidly, these systems face many limitations, creating the need for cloud computing.

Limitations of Traditional IT Systems

1) High Initial Cost

Traditional IT requires heavy upfront investment in servers, storage, networking equipment, software license, and data centers.

2) Limited Scalability

Scaling resources in traditional systems is slow and difficult. New hardware must be purchased and installed when demand increases.

3) Maintenance and Management Overhead

Organizations must manage hardware failures, software updates, security patches, and backups manually.

4) Poor Resource Utilization

Servers often remain underused because they are sized for peak demand.

5) Lack of flexibility and Mobility

Access is usually limited to office networks and fixed systems.

Need for Cloud Computing

1. **Cost Efficiency:** Cloud computing eliminates the need for large upfront investments in hardware and infrastructure. Organizations pay only for

Date _____
Page _____

the resources they use on a subscription or pay-as-you-go basis. This reduces capital expenditure and overall IT costs.

2. Scalability and Elasticity

Cloud resources can be scaled up or down instantly based on demand. This allows organizations to handle workload fluctuations efficiently. It ensures optimal performance without wasting resources.

3. Reduced Maintenance

Cloud service providers manage hardware, software updates, and security. Organizations no longer need to worry about infrastructure maintenance. This allows IT teams to focus on innovation and business growth.

4. Efficient Resource Utilization

Cloud computing uses virtualization and resource pooling to maximize utilization. Resources are dynamically allocated based on user demand. This improves efficiency and reduces wastage.

Q3 Discuss major benefits and applications of cloud computing.

→ Cloud Computing provides on-demand access to computing resources over the internet. It offers several benefits to organizations and individuals and is widely used across different domains.

Major Benefits of Cloud Computing

1. Cost Efficiency.

Cloud computing reduces the need for large upfront investments in hardware and software. Users pay only for the resources they consume, which lowers capital and operational costs. This makes cloud services affordable for startups and small organizations.

2. Scalability and flexibility.

Cloud resources can be easily scaled up or down based on user requirements. This allows organizations to handle varying workloads efficiently.

3. High Availability and Reliability

Cloud providers offer redundancy and backup mechanisms to ensure continuous service availability. Data is stored across multiple servers and locations.

4. Reduced Maintenance and Management

The cloud service provider handles system

Date _____
Page _____

maintenance, software updates, and security patches. This saves time and allows IT teams to focus on core business tasks.

Applications of Cloud computing

1 Data Storage and Backup

Cloud platforms provide secure and scalable storage solutions for large amounts of data. They support automatic backup and easy data recovery.

2 Web Hosting and Application Development

Cloud computing is widely used to host websites and web applications. Developers can build, test and deploy applications using cloud platforms. This reduces development time and infrastructure costs.

3. Business and Enterprise

Organizations use cloud-based applications for email, customer relationship management (CRM), and enterprise resource planning (ERP). These applications improve efficiency and collaboration. Examples include Salesforce and Microsoft 365.

Q4 Describe NIST cloud reference architecture.

→ NIST Cloud Computing Reference Architecture (CCRA) is developed by NIST that defines roles, activities and components for cloud systems. It provides a structured framework defining the basic blocks of cloud computing.

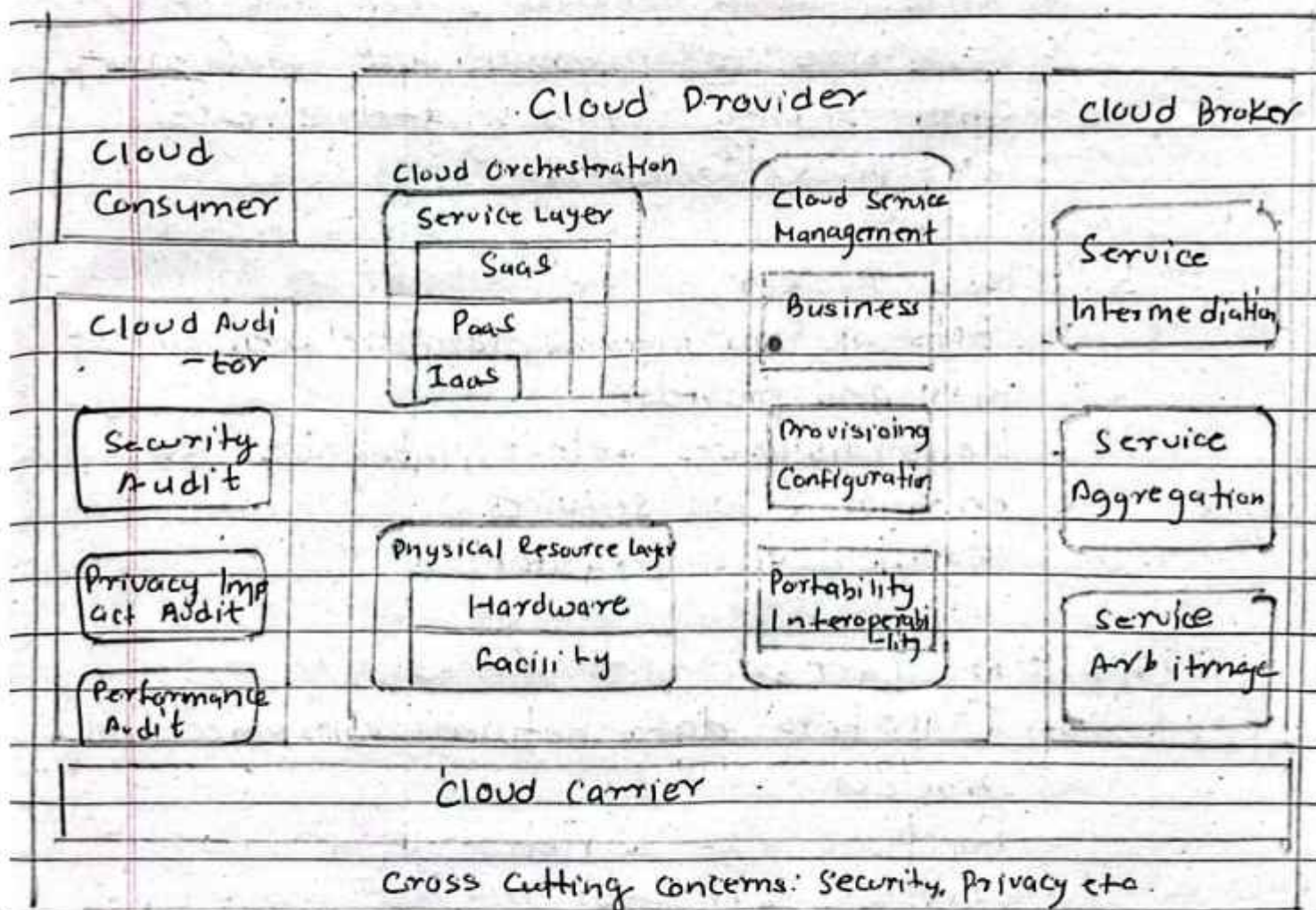


Fig:- Layered View of Cloud Reference Model

Five major actors in NIST cloud computing reference architecture.

1. Cloud Service Consumer

- end user or application using cloud services
- Eg:- End user using Gmail or AWS EC2

2. Cloud Service Provider

- Offers cloud services (IaaS, PaaS, SaaS)
- Responsible for infrastructure, configuration, and security (varies by service type).
- Eg:- AWS, Azure

3. Cloud Broker

- Acts as an intermediary between consumer and provider
- Helps consumers select, integrate, or enhance cloud services.
- Eg:- Cloud aggregator

4. Cloud Carrier

- Transfers data between consumer and provider.
- Functions like a network or internet service distributor
- Eg:- ISP, telecom

3) Cloud Auditor

- Conducts independent assessments of cloud services
- Ensures compliance, security, and performance standards.
- Eg: Third party auditor.

Q5 What are the design principles of cloud architecture? Explain any five.

- Cloud architecture design principles provide guidelines for building cloud systems that are scalable, reliable, secure, and cost-effective.

These principles help organizations fully utilize cloud capabilities while ensuring optimal performance. Any five principles for designing cloud architecture are -

- 1) Scalability
- 2) Resiliency
- 3) Security and Isolation
- 4) Performance
- 5) Cost Optimization

1) Scalability

- Ability to handle growing workloads by adding resources

↳ Horizontal Scaling: Add more machines.

↳ Vertical Scaling: Increase resources of existing machine.

- Date _____
Page _____
- Cloud systems should be designed to scale resources up or down based on demand. This allows applications to handle varying workloads efficiently without performance degradation. Scalability ensures better resource utilization and cost optimization.

2) Resiliency

Fault tolerance:

- System continues operation despite failures
- Ensures the system can withstand failures and recover gracefully.
- Techniques
 - ↳ Redundancy and replication
 - ↳ Load balancing / auto-scaling
 - ↳ Backup and disaster recovery.

Security

3) Security and Multi-Tenancy

Multi-tenant model → shared infrastructure among multiple users

Security Principles:

- Isolation and sandboxing
- Encryption (data-at-rest and in-transit)
- Access Control and identity management (IAM)
- Compliance: GDPR, HIPAA, ISO 27001

4) Performance

- Ensure the application runs efficiently and meets workload demands.
- Configure resources appropriately for performance and scalability.
- Choose suitable instance types, storage options, and networking setups based on workload patterns.

5 Cost-Optimization

- Design the cloud environment to minimize resource wastage.
- Use reserved instances or spot instances when appropriate to reduce costs.
- Continuously monitor usage and spending to identify inefficiencies.
- Avoid over-provisioning; right-size resources according to actual demand.

Q6 Discuss the infrastructural constraints in cloud computing systems.

→ Cloud Computing relies heavily on underlying infrastructure such as networks, data centers, hardware, and virtualization technologies. Despite its advantages, cloud computing faces several infrastructural constraints that can affect performance, reliability, and security.

① Transparency

- Virtualization is key to sharing data and resources in the cloud.
- However, managing multiple servers and data sources makes complete transparency difficult.
- Proper resource visibility, load balancing, and usage reporting are essential to ensure scalability and performance.

② Scalability

- Scaling cloud applications is complex - often requires reconfiguration or re-architecting systems.
- The infrastructure must support elastic provisioning (easy resource allocation and deallocation).
- Systems should be designed to scale dynamically according to demand.

③ Smart Tracking

- To achieve transparency and scalability, intelligent monitoring tools are required.
- Smart tracking ensures continuous observation of performance, resource usage, and system health.
- Helps identify bottlenecks and optimize resource allocation in real time.

④ Security

- The data centers and access nodes (entry points) must be securely designed and protected.
- Security mechanisms should protect both stored and transmitted data.
- Critical to maintain data integrity, confidentiality, and safe access in large distributed environments.

Q7 Explain the evolution of cloud computing from mainframe to cloud platforms.

⇒ The evolution of cloud computing is the result of continuous advancements in computing technology, resource sharing, and networking. It began with centralized mainframe systems and gradually progressed to today's highly scalable cloud platforms.

• 1950s – Mainframes

- Beginning of the automation phase in computing.
- Systems were centralized and used localized infrastructure.
- Access through terminals connected to a single mainframe computer.

- 1960s — Rise of the Personal Computer (PC)
 - Introduction and rising demand for personal desktops.
 - Shift toward decentralized computing
 - Individual users had local processing power
 - IT service industries began to emerge.
- 1990s - Client / server Architecture
 - Development of networked systems connecting clients and servers.
 - Introduction of Virtual Private Networks (VPNs)
 - Growing demand for bandwidth and the rise of the Dot-com revolution (web-based businesses).
- 2000s - Hosted Environment
 - Growth of IT infrastructure outsourcing
 - Widespread adoption of virtualization technology, enabling multiple virtual systems on one physical server
 - Transition from owning hardware to leasing IT infrastructure.

- Beyond 2010 - Cloud Computing Era
 - Emergence of the "as-a-service" model:
 - IaaS (Infrastructure as a Service)
 - PaaS (Platform as a Service)
 - SaaS (Software as a Service)
 - NaaS (Network as a Service)
 - Rise of collaborative and utility computing
 - pay for what you use.
 - Enable scalability, on-demand access, and global connectivity through the internet.

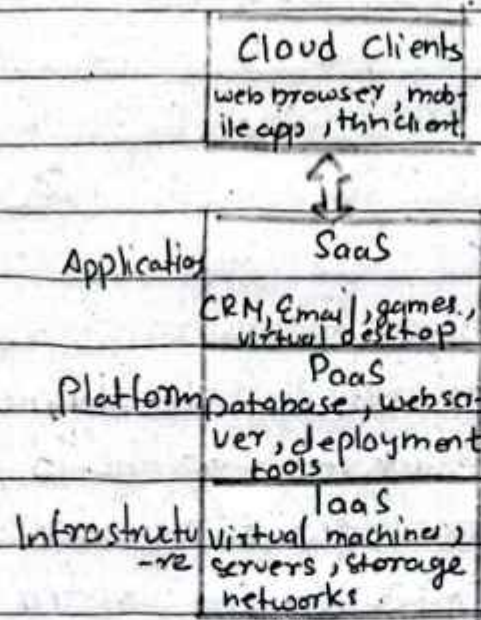
Q8 Describe SaaS, PaaS and IaaS with examples and diagrams.

→ Cloud Services are categorized into three main models, each serving different purposes:

◦ SaaS

◦ PaaS

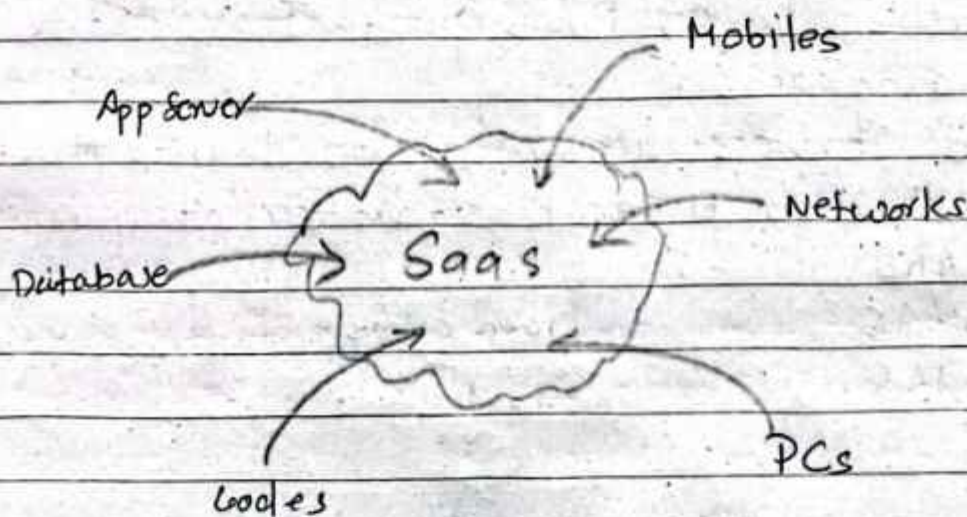
◦ IaaS



1) SaaS : It provides ready-to-use software applications over the internet. User does not manage or control the underlying infrastructure, platform or application code. The service provider handles everything including updates, security and maintenance.

Key Features

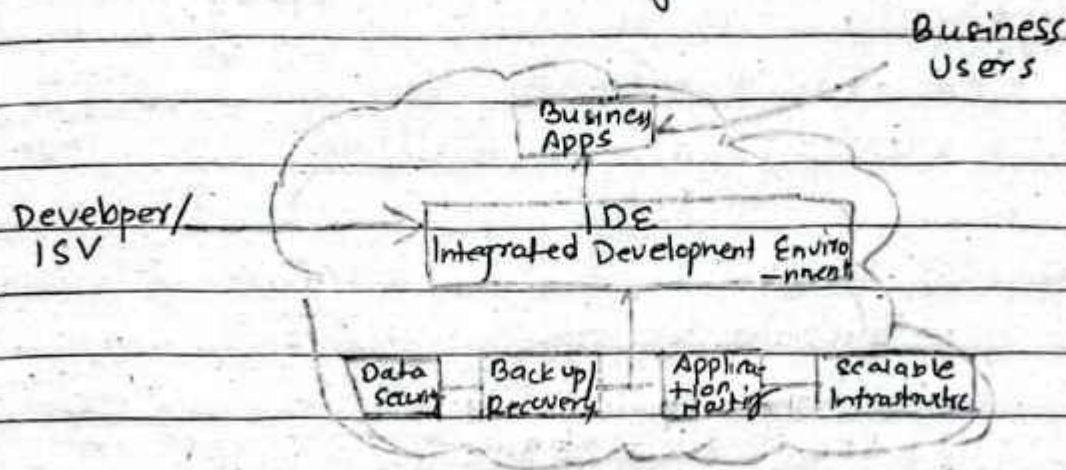
- Accessible through a web browser
- No installation required.
- Subscription or pay as-you-go model



2) PaaS : It provides a development platform that allows developers to build, test and deploy applications. The cloud provider manages infrastructure, operating system, and runtime, while users manage their applications and data.

Key Features

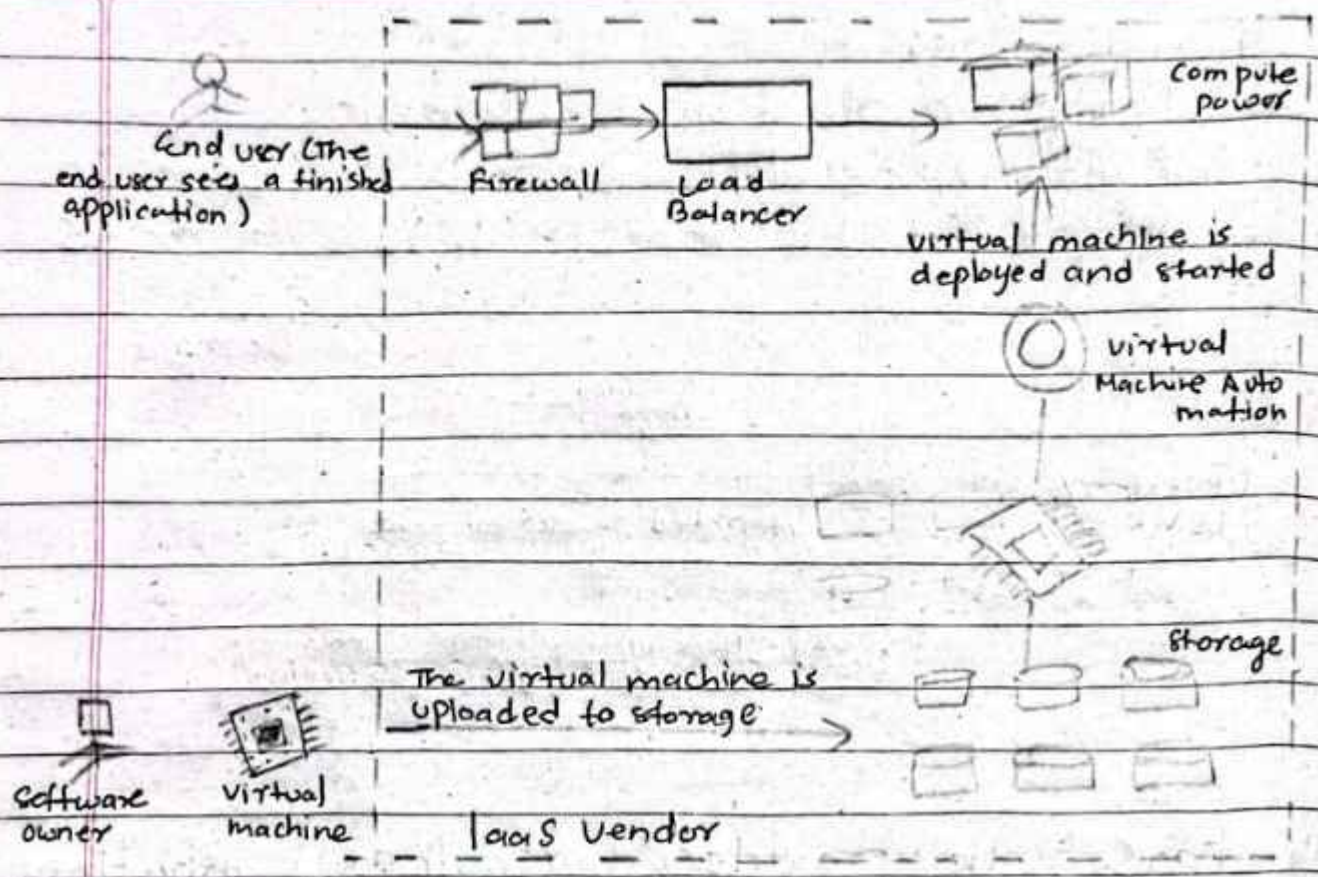
- Faster application development
- Built-in development tools
- No need to manage hardware or OS



3) IaaS : It provides virtualized computing resources such as virtual machines, storage, and networks. Users have full control over operating systems, applications, and configurations, while the provider manages the physical infrastructure.

Key Feature :

- High Flexibility and Control
- Suitable for system administrators
- Pay-per-use model



Q9. If a startup wants low investment and global scalability, should it choose SaaS, PaaS or IaaS? Explain logically.

→ For a startup, the primary concerns are low initial investment, rapid development and the ability to scale globally as the user base grows. Among the three cloud service models - SaaS, PaaS and IaaS - PaaS is the most suitable option to meet these requirements.

Startups generally operate with limited financial and technical resources, making cost efficiency a critical factor. PaaS reduces

initial investment by eliminating the need to purchase, install and maintain physical servers, networking equipment and operating systems.

Since the cloud service provider manages the underlying platform and infrastructure, startup only pay for the resources they actually use, helping them control expenses and avoid unnecessary capital investment.

Global scalability is another important requirement for startups aiming to serve users across different geographical regions. PaaS platforms are built on globally distributed cloud infrastructure and provide automatic scaling features based on application demand. This allows applications to handle increasing user loads seamlessly while maintaining consistent performance without requiring manual resource management.

Furthermore, PaaS accelerates the application development lifecycle by offering ready-to-use development environments, frameworks, databases and tools. Developers can focus on writing and improving application code instead of managing infrastructure-related tasks. This results in faster deployment, reduced development complexity and quicker time-to-market, which is essential for startup growth.

In contrast, SaaS is not suitable for this scenario because it provides ready-made.

Date _____
Page _____

software rather than a platform to develop a startup's own product. IaaS, while offering greater control over infrastructure, requires managing operating systems, security, and system configurations, which increases operational complexity and costs - often unsuitable for early-stage startups.

Q10 An IT company decides to provide free access to a public cloud dedicated to higher education. Which one of the three cloud computing delivery models, SaaS, PaaS or IaaS, should it embrace and why? What applications would be most beneficial for the students? Will this solution have an impact on distance learning? Why or why not?

⇒ If an IT company decides to provide free access to public cloud dedicated to higher education, the most appropriate cloud computing delivery model to embrace is Software as a Service (SaaS).

SaaS is best suited for the education sector because it provides ready-to-use applications that can be accessed through a web browser without requiring users to manage infrastructure or platforms. Students and educators typically lack the technical expertise or need to manage servers, operating systems, or development environments. SaaS eliminates

7

this complexity by allowing users to focus entirely on learning and teaching activities.

Additionally, SaaS has a low entry barrier and works well with limited resources. Since applications are hosted and maintained by the provider, institutions do not need to invest in hardware or software installation. This makes SaaS ideal for offering free and scalable access to a large number of students across different institutions.

Applications Most Beneficial for Students

The following SaaS applications would be most beneficial in a higher-education public cloud:

- Learning Management Systems (LMS) such as Moodle or Canvas for course delivery, assignments and ~~ad~~ assessments.
- Collaboration tools like email, video conferencing, and shared documents for group work and communication.
- Productivity software such as word processors, spreadsheets, and presentation tools for academic tasks.
- Cloud storage services for storing notes, assignments, and research materials.

Impact on Distance Learning

This solution will have a positive impact on distance learning. SaaS applications can be accessed anytime and from anywhere, enabling flexible and remote education. Features such as live classes, recorded lectures, and collaborative tools improve accessibility and learning continuity.

Q11 Explain cloud deployment models: Public, Private, Hybrid & Community.

→ Cloud deployment models define how cloud infrastructure is owned, managed, and accessed. The four main deployment models are Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud.

1. Public Cloud

The public cloud is owned and managed by third-party cloud service providers and made available to the general public over the Internet. Multiple users share the same infrastructure using a multi-tenant model. Examples: AWS, Microsoft Azure, Google Cloud.

Main Benefits of Public Cloud are:

- Easy & Inexpensive Setup
 - Hardware, applications, and bandwidth are managed by the provider.

- Scalability

- Resources can scale to meet changing needs.

- Cost Efficiency

- Pay only for the resources you use; no wasted capacity.

2: Private cloud

A private cloud is dedicated to a single organization and can be hosted either on-premises or by a third-party provider. It offers greater control, security and customization compared to public cloud. Private clouds are commonly used by organizations with strict security and compliance requirements.

Examples :- Private data centers, VMware Cloud

Advantages of Private cloud

- Customer Information Protection

- Sensitive data remains secure within the organization.

- Control and Customization

- infrastructure tailored to organizational needs.

- Compliance

- easier to meet regulatory requirements.

- Performance Optimization

- resources dedicated to internal workloads.

3 a, Hybrid cloud

Hybrid cloud combines both public and private cloud infrastructures, allowing data and applications to move between them. Organizations can keep sensitive data in a private cloud while using public cloud resources for less critical workloads. This model provides flexibility, scalability and cost optimization.

Examples: Private cloud integrated with AWS or Azure.

Benefits

- Improved Security & Privacy
 - Sensitive data stays protected.
- Enhanced Scalability & Flexibility
 - Can handle varying workloads efficiently
- Cost Efficiency
 - Optimizes spending by using public cloud for non-sensitive tasks.

4, Community Cloud

A community cloud is shared by multiple organizations with similar requirements, such as security, compliance, or mission objectives. The infrastructure is jointly used and managed by the participating organizations or a third party. It is commonly used in sectors like education, healthcare and government.

Examples: Government or university shared cloud platforms.

Strengths

- Cost Reduction
 - Shared infrastructures lowers overall expenses
- Improved Security, Privacy & Reliability
 - Maintains standards across members
- Ease of Data Sharing & Collaboration
 - Streamlined for joint projects.

Q12 A multinational company needs data access across continents. Which Cloud deployment model is most suitable and why?

→ For a multinational company that requires data access across continents, the most suitable cloud deployment model is the Hybrid Cloud.

A hybrid cloud combines public and private cloud infrastructures, allowing organizations to leverage the benefits of both. The company can store sensitive or critical data in a private cloud to ensure security and compliance, while using public cloud resources for global access, collaboration and scalability. This ensures that employees across different regions can access data and applications efficiently without compromising security.

Advantages for Multinational Operations

- Global accessibility: Public cloud resources provide high availability and fast access across continents.

- Date: _____
Page: _____
- Security and Compliance :- Sensitive corporate data remains protected in the private cloud.
 - Scalability and Compliance :- Sensitive corporate data remains protected in the private cloud.
 - Scalability :- Resources can be scaled dynamically to meet varying workload demands worldwide.
 - Cost efficiency :- Critical workloads can be optimized between private and public clouds, reducing.

Q13: Discuss Jericho Cloud Cube model with diagrams.

- The Jericho Cloud Cube model is a conceptual framework introduced by the Jericho Forum to analyze security, risk and trust in cloud computing. It helps organizations understand the implications of moving services and data to the cloud by categorizing cloud service based on deployment location, ownership, and control.

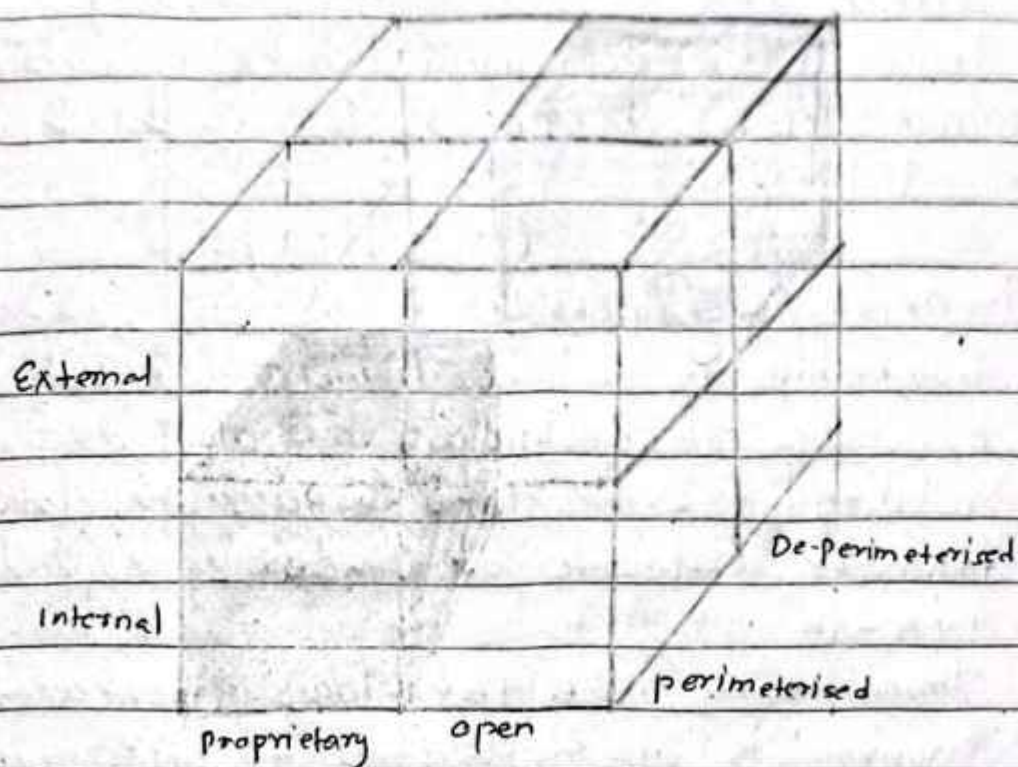


Fig: Cube Showing Classification
(Internal-External, Proprietary-Open, Perimeterised-De-perimeterised)

The model includes four axes along which cloud deployments can be classified:

- 1) Internal / External
- 2) Proprietary / Open
- 3) Perimeterized / De-Perimeterized
- 4) Insourced / Outsourced

1) Internal / External

- The location of data may be internally or externally which ultimately defines the organization's boundary.
- Internal: the cloud services or data reside

Date _____
Page _____

within the organization's boundary .
External : the deployment / data is outside the organization's boundary (e.g, a third-party cloud) .

2, Proprietary / Open

- Ownership is a measurement for not only ownership of technology but also its interoperability, use of data & ease of data-transfer & degree of vendor's application's lock-in
- Proprietary : the technology, interfaces, ownership are restricted to one provider; limited interoperability
- Open : the deployment uses more open standards or multiple suppliers, enabling easier data / partner collaboration .

3, Perimeterized / De-Perimeterized

- Identifies whether traditional network security controls like firewalls are in place or removed / de-perimeterized
- De-perimeterized services require strong encryption, identity management, and access controls

4. In sourced / outsourced

- Sourcing defines whether the customer or the service provider provides the service.
- Insourced: cloud services are provided and managed by the organization's own staff / infrastructure.
- Outsourced: cloud services are delivered via external third-party providers.

Q14 Write a note on the challenges faced in cloud computing.

→ Cloud computing offers many advantages such as scalability, flexibility and cost efficiency. However, it also presents several challenges that organizations must address to ensure effective and secure cloud adoption.

1. Data Security and Privacy

Storing data on the cloud servers raises concerns about unauthorized access, data breaches, and privacy violations. Since data is stored off-premises and often shared across multiple tenants, ensuring confidentiality becomes challenging. Organizations must rely on strong encryption, access controls, and compliance policies.

2, Compliance and Legal Issues

Different countries have different laws regarding data storage and processing. Cloud data may be stored in multiple geographic locations, creating legal and regulatory challenges. Ensuring compliance with standards such as GDPR or data protection laws can be complex.

3 Downtime and Service Availability

Cloud services depend on internet connectivity and provider reliability. Service outages or network failures can disrupt business operations. Even major cloud providers may experience downtime, affecting users worldwide.

4) Vendor Lock-in

Migrating applications and data from one cloud provider to another can be difficult due to differences in platforms and services. This can limit flexibility and increase long-term dependence on a single provider. Vendor lock-in may also raise future costs.

5 Performance and Latency Issues

Application performance can be affected by network latency and bandwidth limitations. Data-intensive or real-time applications may experience delays when hosted in the cloud. Geographic distance from data centers can further impact

response time.

Q15 Explain ethical issues in cloud computing.

→ Cloud Computing enables easy accesses to data and services over the internet, offering convenience and scalability. However, it also raises several ethical issues related to data handling, user rights.

12 1) Data Privacy

Cloud providers store large amounts of user data, often across multiple locations. Users may not know where their data is stored or who can access it. Ethical concerns arise when data is used or shared without proper consent.

2) Data Ownership

Ownership of data in the cloud is sometimes unclear between users and providers. Service agreements may allow providers limited control over stored data. This raises ethical questions about user rights and data misuse.

3) Data Security Responsibility

Security in cloud computing is a shared responsibility between providers and users. Ethical issues occur when accountability for data breaches is unclear. Providers must ensure proper security and transparency.

4, Transparency and Trust

Cloud providers should clearly explain how data is handled and protected. Lack of transparency can reduce user trust. Ethical cloud services require open and honest communication.

5, Compliances and Legal Ethics

Cloud data may be stored in different countries with varying laws. Providers must respect legal and ethical standards across regions. Failures to do so can violate user rights.